

Magitech

TOWERS OF SAEROTH · PLAYER HANDOUT

Machinery built to hold a working, and workings designed around what a machine can do that a hand cannot. Both halves are old; putting them together deliberately is not, and the people doing it have not agreed on what to call it. **Magitech** is the word that has stuck outside the trade, mostly because it is short.

Two ends of the same problem

Melisor Magocracy came at it from the magic. An academy that can already build a teleportation circle wants to know what a circle can be made to do when it is a component rather than a destination, and Melisor has the theorists, the ley-lines and the funding to ask.

Silicar came at it from the machine. A guild that has spent four generations on locks, gearing and irrigation wants to know which of its mechanisms get better with a working inside them, and Silicar has the workshops, the clockwork trade and a habit of building the thing before arguing about it. It is also the guild that invented the firelock, by accident, and has never once let that trade and this one share a bench. See Technology.

Neither of them finds this funny. Everybody else does.

Materials and power

A magitech device is built, not grown out of a spellbook, and it looks it. The good ones use metals and woods that got rare the moment the trade started buying them up, finished the way a luthier finishes a violin rather than the way a smith finishes a plough, and set with a gemstone or crystal more often than not. The metal itself is usually an alloy — a single pure metal holds a working's shape badly, and every workshop guards its own blend like a vintner guards a cellar.

Power is the harder problem, and neither academy solved it from nothing. An aeon stone in this trade is one gem carrying two separate cuts, not a battery wired to a separate working. The **charge rune** comes first and pulls raw magical energy through the gem, and the gem is not an interchangeable vessel for that: some stones catalyse the draw cleanly and some waste half of what the rune is trying to pull through them, which is why a workshop pays for a good ruby and not merely a red one. The **property rune** comes second, into the same stone, and shapes whatever the charge rune drew — and it doesn't take to every gem the same way either. Fire cuts cleanly into ruby and badly into almost anything else; a ward or open air takes to diamond; earth wants peridot. Cut the right rune into the wrong stone and the device is weak or doesn't hold at all, which is most of why the trade's pairings look fixed rather than arbitrary. Twenty years ago nobody in either academy could cut either rune, let alone say which stone suited which. What they had were whole low-level stones pulled out of old ruins, enough of them, studied long enough, to work both cuts back out from the finished article — and to notice, eventually, that the ruins had never once wasted a ruby on anything but fire.

The same ruins gave up broken housings too: alloy and wood still carrying a third cut, **circuitry**, carved into the material itself rather than the stone, that once carried a stone's shaped power the rest of the way into whatever the housing was built to do. That is where the trade got its wiring without inventing it a third time.

Property runes keep turning up in new varieties, and the trade keeps copying or inventing past them — fire, earth, air, a ward, a solvent, a season, with no sign the list is finished. The charge rune never has. Every stone anyone has ever pulled from a ruin carries the same basic cut, and in twenty years nobody has managed to design, or find, a more advanced one. A better stone or a cleaner alloy can close the gap between a device and its theoretical best, but neither can raise the ceiling itself. That ceiling is one specific cut that nobody has learned to improve. Whether that is a limit

of what survived the ruins or a limit neither academy has cracked on its own, Melisor and Silicar give two different answers in public and privately admit to the same one: they don't know.

Not everything that ships is a find. A few pieces are close reconstructions of something pulled half-whole out of a ruin and finished in a workshop; the rest are new work, designed from scratch around the recovered runework and the recovered circuitry, that happened to solve a problem the empire had already solved once. Nobody building either kind knows which one they're holding, because nobody building either kind knows there is a difference to ask about.

The list below names each device by its property rune only — the charge rune riding under it goes unmentioned the way nobody points out the nails in a chair. Twenty years in, Silicar's workshops have shipped a handful of things:

- **The wayfinder** — a disc small enough to wear on a cord, pointing toward whatever its stone was keyed to find. The one every court now wants and almost none can buy; Silicar's guild fills perhaps thirty orders a year.
- **The cold lamp** — no flame, no oil, no working laid on fresh each evening. A stone behind hammered glass, burning level and cold until it runs dry.
- **The strike-stone** — a ruby cut with a fire rune, set into a case of alloy the size of a thumb. Click it and it lights. No oil, no flint, no wick, which makes it the one magitech item an ordinary household might actually own, on the rare year it can find one to buy.
- **The slate** — a peridot cut with an earth rune, bedded into a tablet of worked stone. Write on it, press the rune, and the mark is gone — a scribe's whole complaint against parchment, solved. Thurion Merchant Alliance's counting-houses are already trying to buy the patent out from under Silicar before a rival house does.
- **The diving mask** — a diamond cut with an air rune, set into the glass. A salvage crew wearing one can work a wreck for the better part of an hour on a single charge, which has already changed what a reef-city considers worth the dive.

Every one of them is rare, every one is expensive, and every one works exactly as many times as its stones allow, and not once more.

Charge, pattern, impedance

Every apprentice guesses the same thing in their first week at a bench: that the stone, the cut and the metal all multiply together, and more of any one makes a better device. Three things do matter. It isn't the word *times* — though the shorthand chalked over every bench in Brightfurrow looks close enough to it:

$$W = Q \cdot P \div Z$$

Charge (Q) comes from the charge rune, but the gem it's cut into still matters. The rune sets a firm ceiling — no cut anyone has recovered pulls more than a low-level draw — and the gem decides how close to that ceiling a given stone actually gets. Some stones catalyse cleanly and some waste half of what the rune is trying to pull through them, which is why a clean ruby costs more than a merely red one even cut with the identical rune. Every charge rune anybody has ever recovered from a ruin is the same basic cut, and in twenty years nobody has managed to design a more advanced one. The ceiling on the whole trade is this one rune. What a good stone buys is getting near it.

Pattern (P) is the property rune, cut second into the same stone, and it has a ceiling of its own: how well it takes to the gem underneath it. Fire cuts cleanly into ruby and badly into almost anything else; a ward or open air takes to diamond; earth wants peridot. Cut a rune into a stone with no affinity for it and the device is weak or doesn't hold at all, which is most of why the trade's pairings look fixed rather than arbitrary. Because the cut is made once, when the stone itself is made, there is no reworking it after. Nobody reprograms an aeon stone. They requisition a new one, in the right gem this time, and cut it again.

Impedance (Z) is the circuitry — a third cut, carved not into the stone but into the alloy or the wood that houses it, carrying the shaped power the rest of the way out and refining it again on the way. Charge and Pattern both have a ceiling a good stone can approach but never clear. Impedance has none. Nobody has ever worked out a limit to how good a circuit can get, which is the real reason every guild in Brightfurrow guards its own circuitry patterns and alloy blends rather than publishing either. Bad impedance doesn't just waste charge as heat and backlash, burning through the stone in a third the uses a well-carved circuit would — it refines the effect badly too, which is the difference between a warded clasp that holds a clean half-shield and one that flickers.

That chalked-up version is a simplification, and it earns its keep at a bench where nobody has time for better. It treats one activation as a single lump of charge spent all at once, which is not what actually happens inside the housing. What happens is a stone bleeding charge over the fraction of a second the working takes, and a rate is not a lump:

$$w(t) = -(dQ/dt) \cdot P(t) \div Z(t)$$

The device's output at any instant is however fast the stone is losing charge right then, shaped by the pattern and cut down by whatever is in the way. None of that changes what a stone can afford across its whole working life, because a rate integrates back into a total the same way it always has:

$$\int_0^T w(t) dt \leq Q(0)$$

Spend it fast in one dramatic burst, or slow across a hundred small pulls, and the total is bounded the same either way. That is the actual shape of "works exactly as many times as its stone allows, and not once more" — not a count of uses at all, but an area under a curve that happens, for a well-made device with a steady draw, to look enough like a count that nobody in the trade has ever bothered to say it differently.

Put the three variables together, chalked or worked in full, and the trade's whole twenty years comes down to one number nobody can move, one number nobody can invent past, and one number that is the entire difference between a wayfinder and a toy. Melisor Magocracy has never had a reason to chase the third one. A warded clasp cut at Thelemar runs dry in a fraction of the uses a Silicar lamp gets from the same grade of stone, and nobody on the Council has ever once complained about it.

Prestige pieces

Melisor Magocracy never needed magitech to solve anything. An academy that can put three teleportation circles under one campus has never once lacked a practical answer, so its half of the trade was never aimed at a problem. It was aimed at whoever already has every option and wants to be envied for the one they picked.

- **The warded clasp** — a diamond cut with an abjuration rune, pinned at the collar. It raises the same half-shield a first-year apprentice learns to cast, and holds it just as well, which has made it wildly popular with nobles who want to duel without earning the privilege the ordinary way. Ashwin Devaraj, who has spent a career being paid for that privilege and cited for none of it, has opinions about the clasp and states them loudly, usually in a tavern.
- **The presser** — a topaz cut with a cleansing rune, set into a flat iron. A launderer does the same work for a fraction of the price and cannot be shown off at a dinner party, which is rather the point of owning one instead.
- **The seasoning dish** — a citrine cut with a transmutation rune, laid under the centrepiece at a Council banquet. No two academies season a dish the same way, and a Thelemar dinner argues its politics through the courses as much as through anyone's conversation.

None of it is cheaper than paying somebody to do the job. That was never the offer.

The stakes

- **It is the first field in a century where Melisor Magocracy is not obviously ahead.** Silicar is small, is not a great power, and is producing results the academies did not predict.
- **The Council of Archmages cannot rule on it,** because two academies claim the same process and the Council's horror of ruling on anything is a policy older than the dispute.
- **Everything it produces is unlicensed,** in the same way the firelock is unlicensed. See Technology.
- **It is expensive and it is visible.** A magitech workshop is a large building with a great deal of money in it and no guild old enough to have arranged its own protection.

You cannot repeat any of this. Not to the party, not on paper, not in stone.